

Project: Hydrogen Compressor Upgrade

Document Type: Technical Specification

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1. Scope

This document defines requirements for upgrading the hydrogen compressor system in Unit B.

The target is to improve availability, reduce vibration trips, and increase flow stability.

2. Design Requirements

- Rated flow: 12,000 Nm³/h
- Suction pressure: 18 bar
- Discharge pressure: 42 bar
- Max vibration velocity: 4.5 mm/s RMS
- Bearing temperature alarm: 85 C
- Bearing temperature trip: 95 C

3. Instrumentation

- Vibration probes on DE and NDE bearings
- Redundant pressure transmitters at suction and discharge
- RTDs for bearing and lube oil temperatures
- Flow meter with 1-second sampling interval

4. Risk and Compliance

- Risk: Seal gas contamination may cause premature wear.

- Mitigation: Install dual-stage filtration and weekly gas quality checks.
- Compliance: IEC 60034 and plant safety standard PS-118 shall be followed.
- Cybersecurity controls for PLC communication must align with internal policy.

5. Commissioning

- Pre-start checks for lube oil pressure, cooling water flow, and interlock logic.
- Perform no-load test for 2 hours and full-load test for 8 hours.
- Acceptance criteria: no trips, vibration < 4.5 mm/s, stable discharge pressure.

6. Maintenance Strategy

- Monthly condition monitoring
- Quarterly alignment verification
- Annual overhaul planning based on vibration trend and oil analysis